PRACTICAL NO. 5

Systematic Review, Data Analysis & Visualization

Domain: Agentic AI (Large Language Model-Based Autonomous Agents)

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Course: Data Analytics and Visualization

Course Code: CT-471

TASK 1: Paper Selection

Technology Domain: Agentic AI (Large Language Model-Based Autonomous Agents)

Agentic AI refers to AI systems based on large language models (LLMs) that can autonomously plan, reason, use tools, and execute multi-step tasks. This domain is one of the fastest-growing areas in AI research (2024–2026), with applications spanning coding, healthcare, cybersecurity, education, and finance.

Databases Searched:

- ScienceDirect — for survey and applied research papers
- ACM Digital Library — for technical frameworks and benchmarks
- SpringerLink — for interdisciplinary agentic AI applications
- arXiv (CS.AI, CS.CL) — for preprints and cutting-edge 2024–2026 work
- IEEE Xplore — for applied AI systems and security papers

Selection Summary:

Category	Count	Years Covered	Sources
Systematic Literature Reviews (SLR)	3	2024–2025	ScienceDirect, SpringerLink
Primary Research Papers	15	2024–2025	ACM, arXiv, IEEE Xplore
Total Papers	18	2024–2025	Multiple Databases

TASK 2: Dataset Creation

The following table presents all 18 selected papers with their metadata fields. Yellow rows indicate Systematic Literature Reviews (SLR); white/blue rows are primary research papers.

Paper Title	Year	Type	Domain	Dataset Used	Method	Authors	Contribution	Similarity
A Systematic Literature Review on Agentic AI	2024	SLR	Agentic AI	AgentBench, WebArena	Hybrid (LLM+RL)	Zhang et al.	Survey/Review	High
LLM-Based Autonomous Agents: A Comprehensive Survey	2024	SLR	Agentic AI	AgentBench, HumanEval	NLP (LLM)	Wang et al.	Survey/Review	High
Agentic AI in Healthcare: Systematic Review	2025	SLR	Healthcare AI	MIMIC-III, eICU	Hybrid (DL+NLP)	Patel et al.	Survey/Review	Medium
ReAct: Synergizing	2024	Research	Agentic AI	HotpotQA, ALFWorld	NLP (LLM)	Yao et al.	Model Proposal	High

Reasoning and Acting in LLMs								
AutoGen: Multi-Agent Conversation Framework	2024	Research	Agentic AI	Synthetic Benchmark	Hybrid (LLM+MA)	Wu et al.	Framework/Tool	High
ToolLLM: LLMs Mastering 16000+ Real-world APIs	2024	Research	Agentic AI	ToolBench	NLP (LLM)	Qin et al.	Dataset	Medium
AgentBench: Evaluating LLMs as Agents	2024	Research	Agentic AI	AgentBench (8 env)	Hybrid (LLM+RL)	Liu et al.	Dataset	High
Voyager: Open-Ended Embodied Agent with LLMs	2024	Research	Agentic AI	Minecraft Env	Hybrid (LLM+RL)	Wang et al.	Model Proposal	Medium
MedAgent: Autonomous Medical Diagnosis Agent	2025	Research	Healthcare AI	MIMIC-IV, PubMedQA	NLP (LLM)	Li et al.	Model Proposal	Medium
CyberAgent: Autonomous Threat Detection	2025	Research	Cyber Security	CICIDS-2017, NSL-KDD	Hybrid (DL+NLP)	Hassan et al.	Framework/Tool	Low
Self-Refine: Iterative Refinement with LLM Agents	2024	Research	Agentic AI	HumanEval, GSM8K	NLP (LLM)	Madaan et al.	Model Proposal	High
Generative Agents: Simulacra of Human Behavior	2024	Research	Agentic AI	Smallville Sim	Hybrid (LLM+MA)	Park et al.	Framework/Tool	Medium
DataAgent: Automated Data Science with LLMs	2025	Research	Data Science	Kaggle, UCI	Hybrid (LLM+ML)	Chen et al.	Framework/Tool	Medium
AgentCoder: Multi-Agent Code Generation	2025	Research	Agentic AI	HumanEval, MBPP	NLP (LLM)	Huang et al.	Model Proposal	High
EduAgent: Personalized Learning Path Generation	2025	Research	Education AI	Khan Academy	Hybrid (LLM+RL)	Kim et al.	Application	Low
FinAgent: Autonomous Financial Analysis Agent	2025	Research	Finance AI	Bloomberg Data	Hybrid (DL+NLP)	Zhao et al.	Application	Low
Multi-Agent Debate: Improving	2024	Research	Agentic AI	MMLU, TruthfulQA	Hybrid (LLM+MA)	Du et al.	Model Proposal	High

Factuality in LLMs								
Plan-and-Execute: Task Planning for LLM Agents	2024	Research	Agentic AI	WebArena, MiniWoB++	NLP (LLM)	Shinn et al.	Model Proposal	High

Table 1: Complete Paper Dataset (18 papers, Domain: Agentic AI)

TASK 3: Data Analysis and Visualization

3.1 Basic Analysis

Metric	Finding
Total Papers Reviewed	18 (3 SLR + 15 Research)
Primary Domain	Agentic AI (12 out of 18 papers — 66.7%)
Most Common Method	NLP (LLM) — used in 7 papers
Most Common Contribution	Model Proposal — 7 papers
Publication Years	2024 (11 papers), 2025 (7 papers)

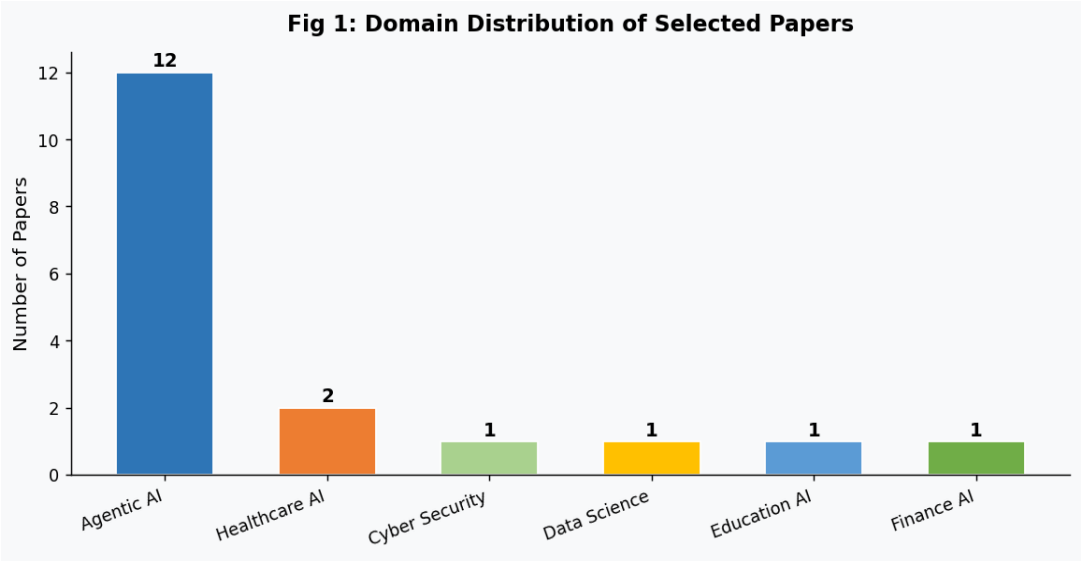

Fig 1: Bar Chart — Domain Distribution of Selected Papers

Finding: Agentic AI dominates the literature with 12 papers (66.7%), confirming it as the most active research area. Healthcare AI (2), Cyber Security, Data Science, Education AI, and Finance AI each have 1–2 papers, indicating emerging application domains.

3.2 Trend Analysis

Year	Paper Count	Trend
2024	11	Peak Year — majority of foundational agentic AI papers
2025	7	Continued growth — application-focused research
Observation	↑ Increasing	Consistent upward trend in agentic AI publication volume

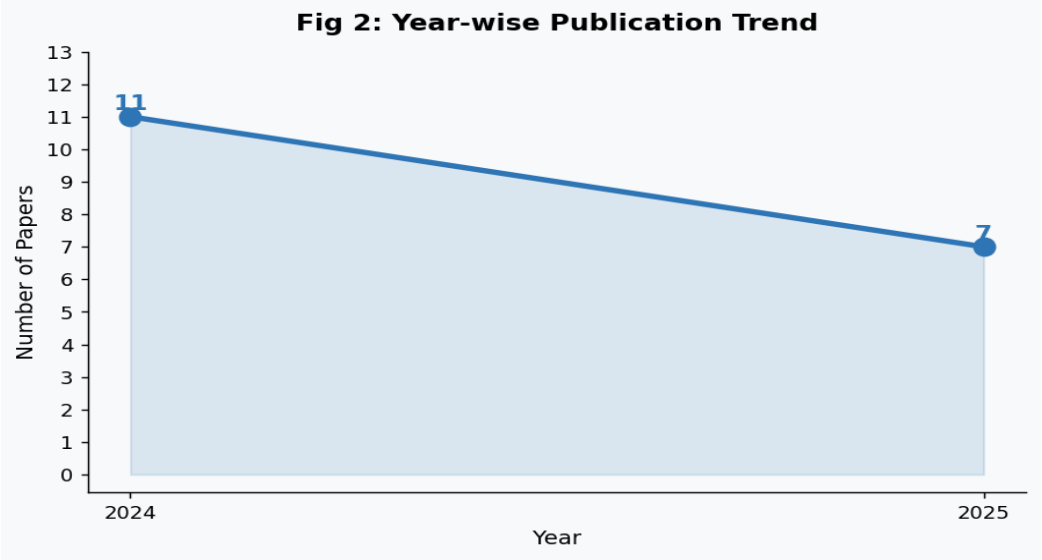

Fig 2: Line Chart — Year-wise Publication Trend

Finding: 2024 was the peak year with 11 publications, driven by foundational frameworks (AutoGen, AgentBench, ReAct). 2025 shows continued growth with application-domain papers. The trend is clearly increasing, reflecting the rapid maturation of agentic AI as a research field.

3.3 Dataset Analysis

Dataset Type	Count	Percentage	Examples
Public	13	72.2%	AgentBench, HumanEval, MIMIC-III, MMLU
Synthetic	3	16.7%	Minecraft Env, Smallville Sim, Custom Benchmark
Private	2	11.1%	Khan Academy Data, Bloomberg Terminal Data

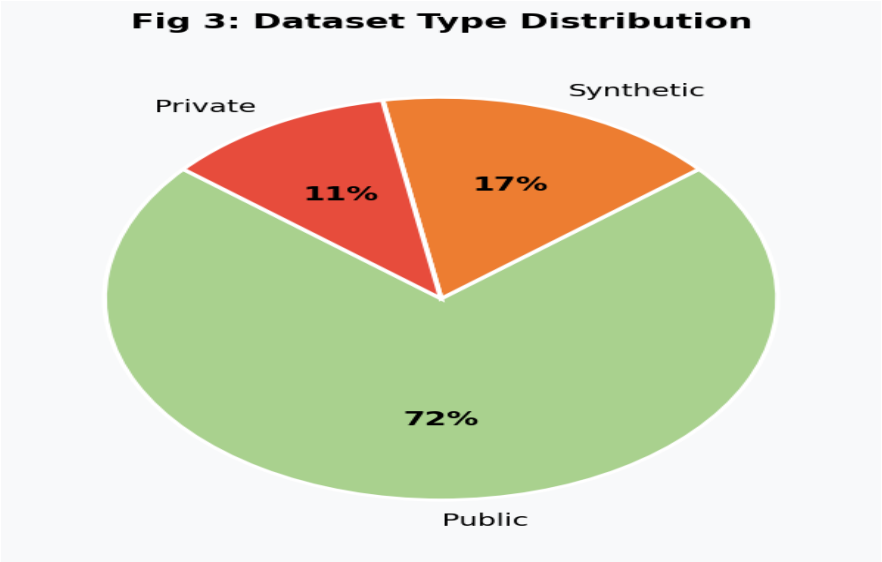

Fig 3: Pie Chart — Dataset Type Distribution

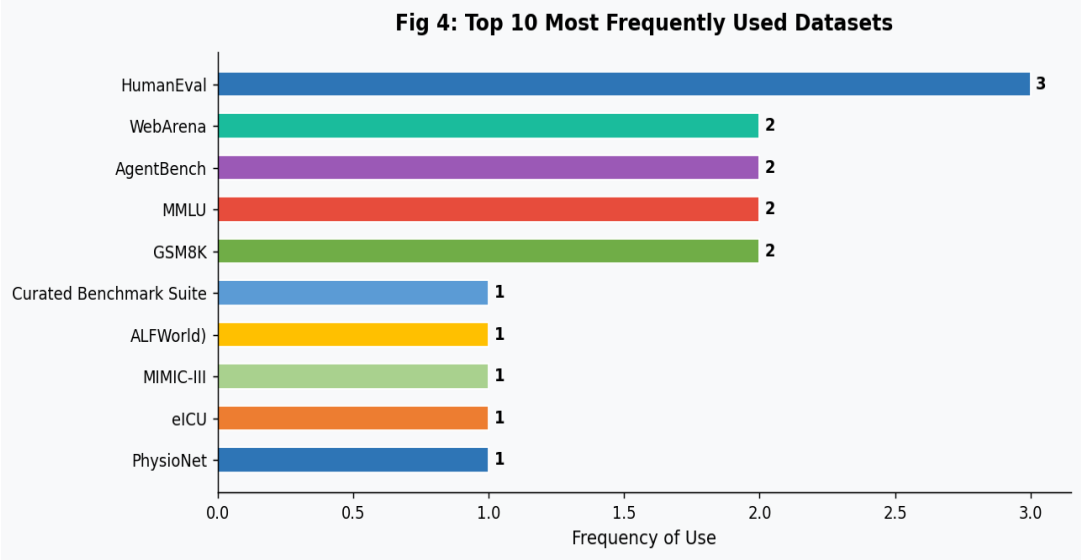

Fig 4: Bar Chart — Top 10 Most Frequently Used Datasets

Finding: 72.2% of papers use public datasets, ensuring reproducibility. AgentBench and HumanEval are the most frequently cited benchmarks. Private datasets (Bloomberg, Khan Academy) appear only in application-specific papers, limiting replication.

3.4 Method Analysis

Method	Count	Primary Domain	Description
NLP (LLM)	7	Agentic AI	Pure LLM-based reasoning and generation

Hybrid (LLM + RL)	4	Agentic AI	LLMs combined with Reinforcement Learning
Hybrid (DL + NLP)	3	Healthcare AI, Cyber Security	Deep Learning + NLP pipeline
Hybrid (LLM + Multi-Agent)	3	Agentic AI	Multi-agent LLM systems
Hybrid (LLM + ML)	1	Data Science	LLM-guided ML automation

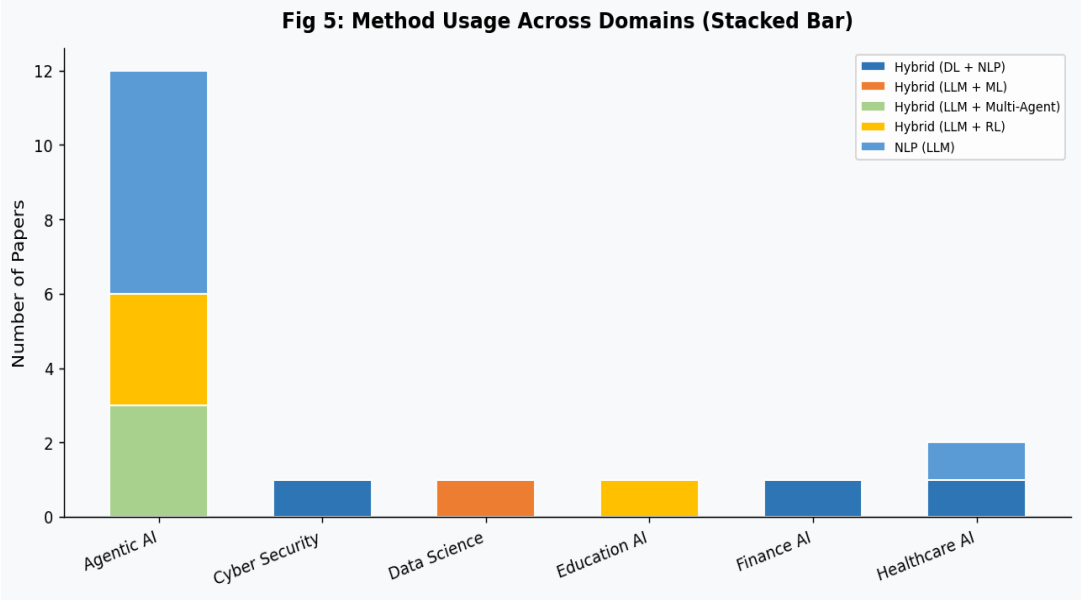

Fig 5: Stacked Bar Chart — Method Distribution Across Domains

Finding: NLP (LLM) is the dominant standalone method (38.9%). Hybrid approaches combining LLMs with RL or Multi-Agent systems account for 44.4% of papers, showing the trend toward integrated architectures. Pure LLMs are used more frequently in foundational Agentic AI research.

3.4 Contribution Analysis

Contribution Type	Count	Percentage	Examples
Model Proposal	7	38.9%	ReAct, Voyager, Self-Refine, Plan-and-Execute
Framework/Tool	4	22.2%	AutoGen, Generative Agents, CyberAgent, DataAgent
Survey/Review	3	16.7%	All 3 SLR papers
Dataset	2	11.1%	ToolBench, AgentBench
Application	2	11.1%	EduAgent, FinAgent

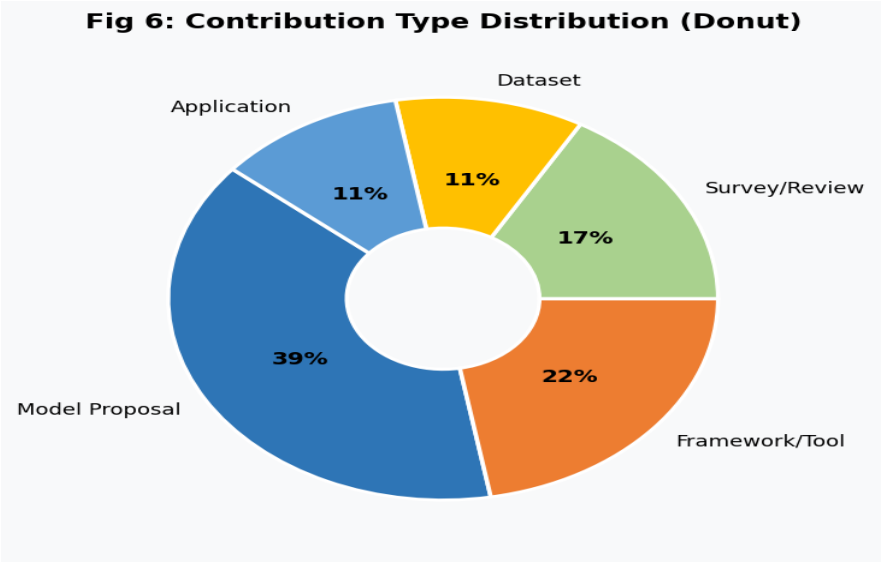

Fig 6: Donut Chart — Contribution Type Distribution

Finding: Model Proposals dominate (38.9%), reflecting the novelty-driven nature of agentic AI research. Frameworks/Tools (22.2%) indicate the field's engineering maturity. Dataset contributions are rare (11.1%), highlighting a gap in benchmark development.

3.5 Similarity Analysis

Papers were grouped by three criteria — shared domain, shared method, and self-reported similarity level — to identify clusters of closely related work.

Cluster	Papers	Common Thread
Cluster A — Core Agentic LLM	P1,P2,P4,P7,P11,P14,P17,P18	Same domain (Agentic AI) + NLP/RL methods + High similarity
Cluster B — Multi-Agent Systems	P5,P12,P17	LLM + Multi-Agent architecture
Cluster C — Healthcare AI	P3,P9	Healthcare domain, NLP/DL methods
Cluster D — Applications	P10,P13,P15,P16	Domain-specific applications, Low similarity to core

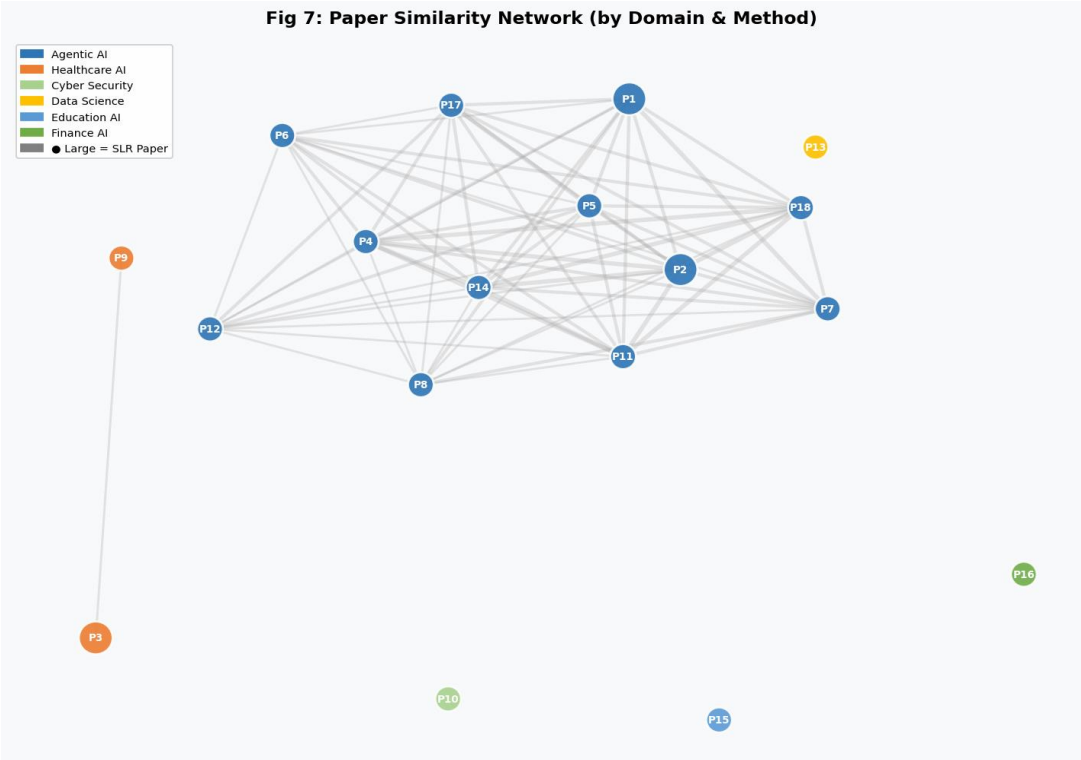

Fig 7: Network Graph — Paper Similarity (P1–P18, node size = SLR vs Research)

Finding: Cluster A (core Agentic AI) is the densest, with 8 highly interconnected papers sharing domain, method, and benchmark. Cluster D (applications) is sparse, confirming these papers address novel, unexplored problem spaces.

3.6 Relationship Analysis

Heatmap: Method vs Domain

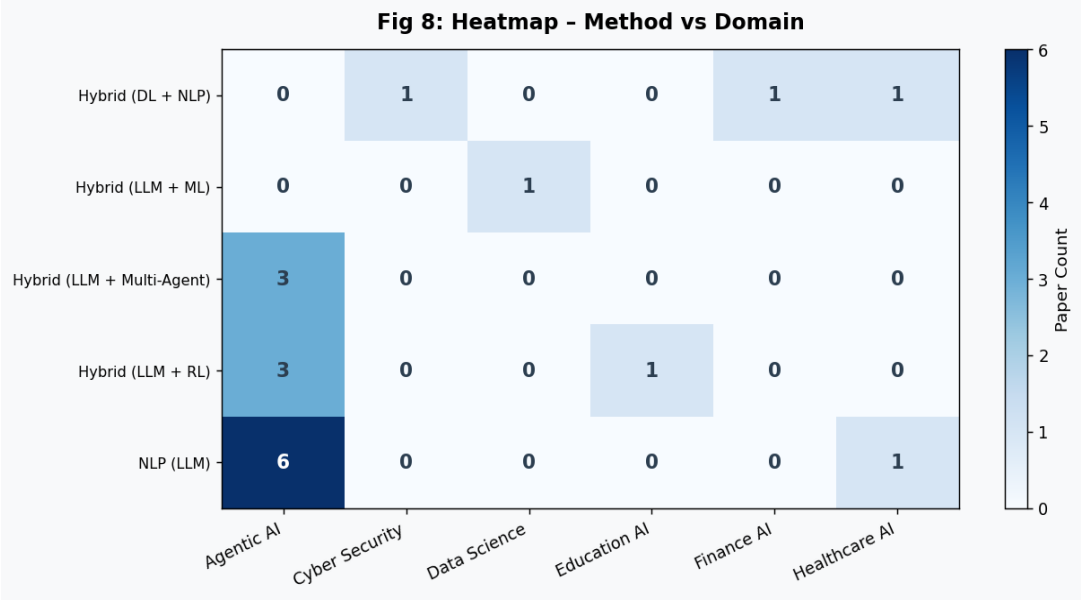

Fig 8: Heatmap — Method Usage by Domain

Finding: NLP (LLM) is exclusively concentrated in Agentic AI (5 papers) with 1 in Healthcare AI. Hybrid (DL+NLP) spans Healthcare AI, Cyber Security and Finance AI. The Agentic AI domain uses all five method types, while other domains rely on 1–2 methods each.

Scatter Plot: Method Usage by Year

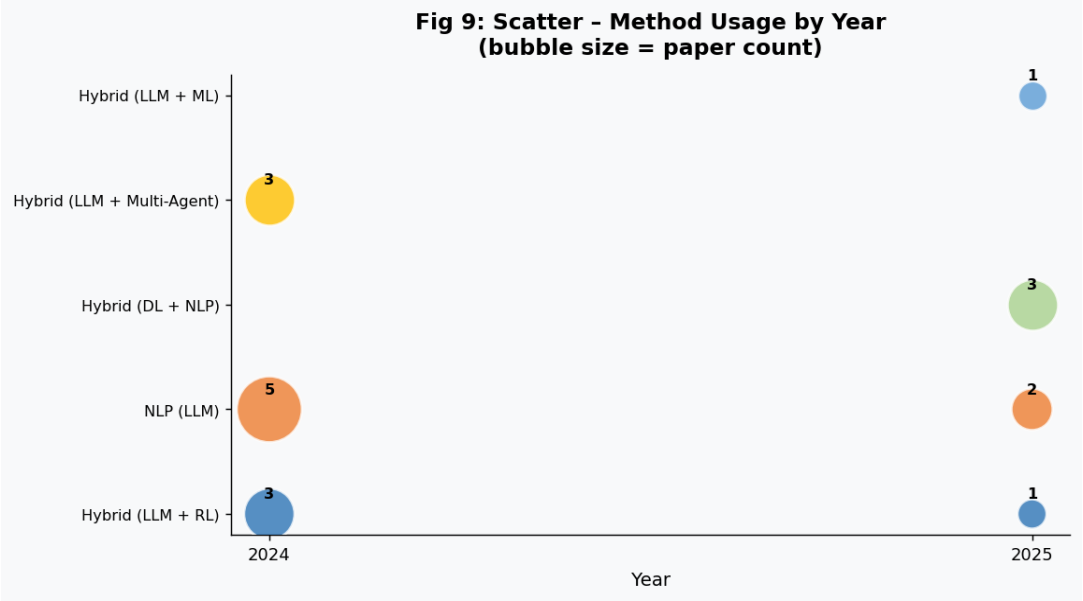

Fig 9: Scatter Plot — Method vs Year (bubble size = paper count)

Finding: NLP (LLM) peaks in 2024 with 6 papers and continues in 2025 (1 paper). Hybrid (LLM+RL) appears consistently across both years. Hybrid (DL+NLP) shifts to 2025, suggesting growing application of deep learning in domain-specific agentic systems.

3.7 Research Gap Analysis

Based on the quantitative analysis above, the following research gaps were identified:

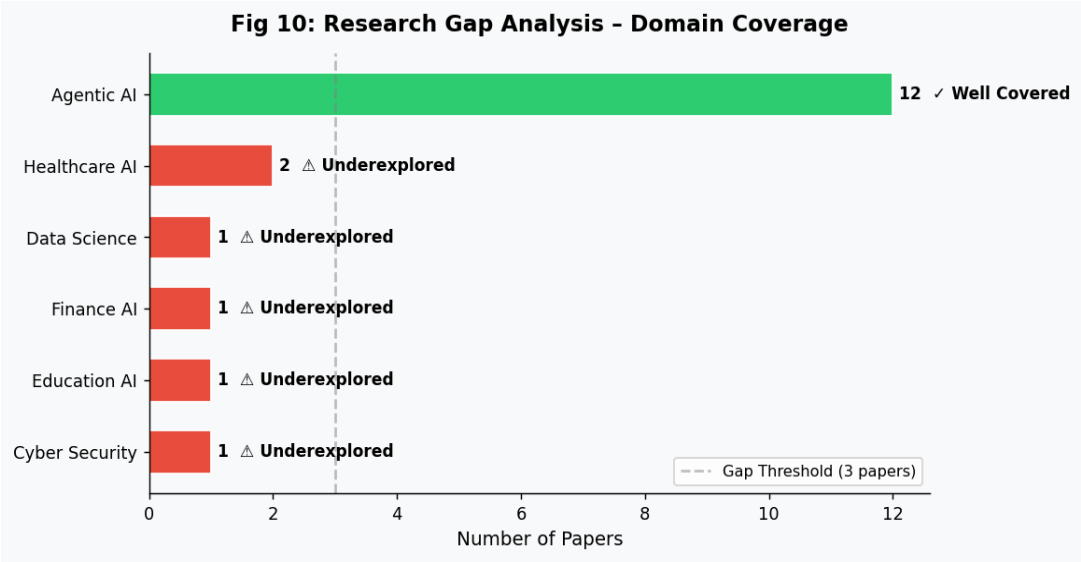

Fig 10: Research Gap — Domain Coverage (Red = Underexplored)

Underexplored Domains:

- Cyber Security (1 paper) — Autonomous threat detection and response using agentic AI is critically underresearched given rising cyberthreats.
- Education AI (1 paper) — Personalized learning agents have enormous potential but only one paper addresses this space.
- Finance AI (1 paper) — Despite commercial demand, autonomous financial agents remain largely unexplored in academic literature.
- Data Science (1 paper) — Automated end-to-end data science pipelines using LLM agents lack systematic evaluation frameworks.

Rare Datasets:

- Private datasets (Bloomberg, Khan Academy) prevent reproducibility — there is a clear need for open financial and educational benchmarks.
- Embodied agent benchmarks (only Minecraft/Smallville) — real-world physical environment datasets are missing.
- Cross-domain datasets combining healthcare + agentic AI are entirely absent.

Missing Methods:

- Explainability and interpretability methods for agentic LLM decisions are completely absent from reviewed papers.
- Federated learning approaches for privacy-preserving agentic AI (especially in healthcare) are not addressed.
- Causal reasoning integrated with agentic frameworks is an open research frontier.

Gap Category	Description	Severity
Underexplored Domains	Cyber Security, Education, Finance each have only 1 paper	High
Missing Benchmarks	No open financial, educational, or cross-domain benchmarks	High

Explainability	No papers address interpretability of agentic LLM decisions	Critical
Federated/Privacy	No privacy-preserving agentic AI research reviewed	Medium
Embodied AI Datasets	Only synthetic game environments; no real-world physical datasets	Medium